

Randomized and Unbounded LP

- 1 Duality or Polarity
 - polygons defined by half planes
- 2 Randomized Linear Programming
 - shuffle the input
 - expected running time
- 3 Unbounded Linear Programs
 - inward pointing normals
 - reduction to one dimension
- 4 the LP solver of CGAL
 - a first example

MCS 481 Lecture 12
Computational Geometry
Jan Verschelde, 21 September 2026

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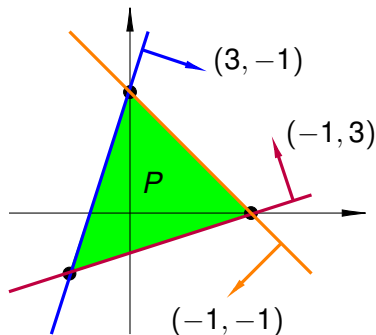
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polygons defined by half planes

P is a polygon, defined by a finite set of half planes:

$$\text{for all } (x_1, x_2) \in P: v_1 x_1 + v_2 x_2 \geq b, \quad b = \min_{(x_1, x_2) \in P} v_1 x_1 + v_2 x_2,$$

where (v_1, v_2) is an inward pointing normal, b is the offset.



P is defined by

$$(-1)x_1 + (-1)x_2 \geq -2$$

$$(-1)x_1 + 3x_2 \geq -2$$

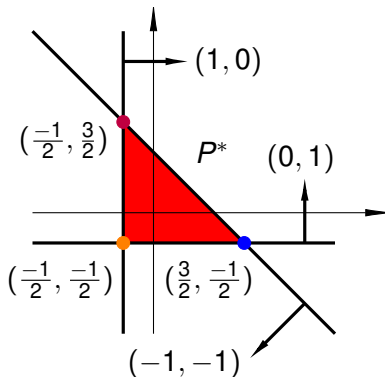
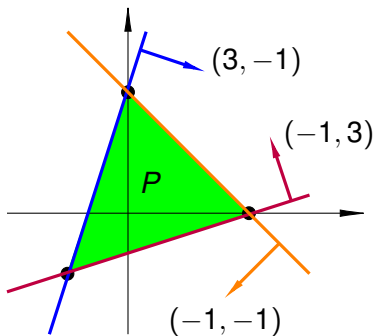
$$3x_1 + (-1)x_2 \geq -2$$

duality or polarity

P is defined by a finite set of half planes:

$$(-1)x_1 + (-1)x_2 \geq -2, \quad (-1)x_1 + 3x_2 \geq -2, \quad 3x_1 + (-1)x_2 \geq -2.$$

Let $(-1/2, -1/2)$, $(-1/2, 3/2)$, $(3/2, -1/2)$ as $(v_1/(-b), v_2/(-b))$, associated with (v_1, v_2) and b . The dual of P is P^* .



some exercises

Exercise 1: What is the condition on P for the construction of P^* ? Is this a condition that always can be made to satisfy?

Exercise 2: Define the dual for the example on the previous slides using *outward* pointing normals.

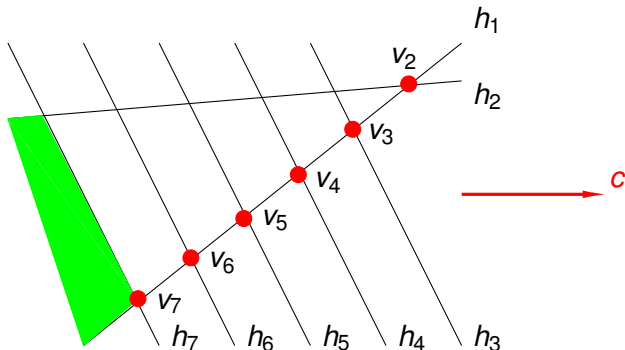
Exercise 3: Given P as the convex hull of a finite set of points, define the algorithm to construct P^* . Illustrate your algorithm on a random polygon with 10 points.

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an unfortunate order of the input

Given are $H = (h_1, h_2, \dots, h_7)$ and the objective c .



Because $v_{i-1} \notin h_i$, we have the worst case $O(n^2)$ cost.

shuffle the input

A randomized version shuffles the order of the input.

Algorithm 2DRANDOMIZEDBOUNDEDLP(H, c)

Input: $H = (h_1, h_2, \dots, h_n)$, $c = (c_1, c_2)$.

Output: (x, y) optimal vertex or i if $P_i = \emptyset$.

- 1 Let p be a random permutation on n elements.
- 2 Let pH be H , permuted with p .
- 3 Return 2DBOUNDEDLP(pH, c).

There are $n!$ possible orders, what is the expected cost?

The expected cost is the average cost over all $n!$ permutations.

an algorithm for a random permutation

Exercise 4:

- 1 Formulate an algorithm to generate a random permutation.
- 2 Assuming the cost of generating a random number is $O(1)$, show that the cost of generating a random permutation on n elements is $O(n)$.

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expected running time

Let us analyze the expected running time. Consider

$$X_i = \text{random variable} \in \{0, 1\} = \begin{cases} 0 & \text{if } v_{i-1} \in h_i \\ 1 & \text{if } v_i \notin h_i \end{cases}$$

then the cost is $\sum_{i=1}^n O(i)X_i$.

The expected cost is $\sum_{i=1}^n O(i)E(X_i)$,

where $E(X_i)$ is the probability that h_i is one of the half planes in $H = (h_1, h_2, \dots, h_n)$ which goes **not** through v_{i-1} .

If $v_{i-1} \notin h_i$, then we need $O(i)$ steps for the 1D LP.

If $v_{i-1} \in h_i$, then v_i lies on ℓ , the line bounding h_i .

backward analysis

Lemma (probability of a half plane passing through vertex)

Let $X_i \in \{0, 1\}$ be a random variable: $X_i = 0$ if $v_{i-1} \in h_i$ and $X_i = 1$ if $v_{i-1} \notin h_i$. Then $E(X_i) \leq 2/i$.

Proof. At the end of the algorithm $i = n$, what is the probability that $v_{n-1} \notin h_n$? At least two hyperplanes pass through v_n .

If we choose twice from $H = (h_1, h_2, \dots, h_n)$, then the probability to select h_n is at most $2/n$.

Now think backwards and consider $(h_1, h_2, \dots, h_i)$. Then the probability that h_i passes through v_i is at most $2/i$. Thus $E(X_i) \leq 2/i$. Q.E.D.

Theorem (cost of randomized 2D LP)

The cost of 2DRANDOMIZEDBOUNDEDLP is $O(n)$.

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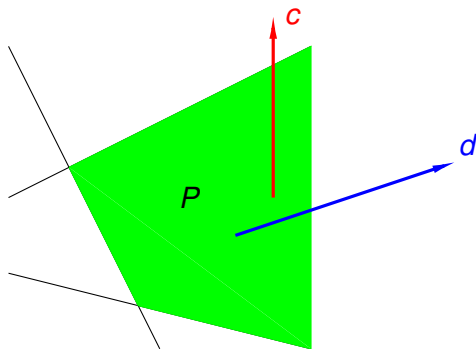
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unbounded linear programs

An LP program is *unbounded* if

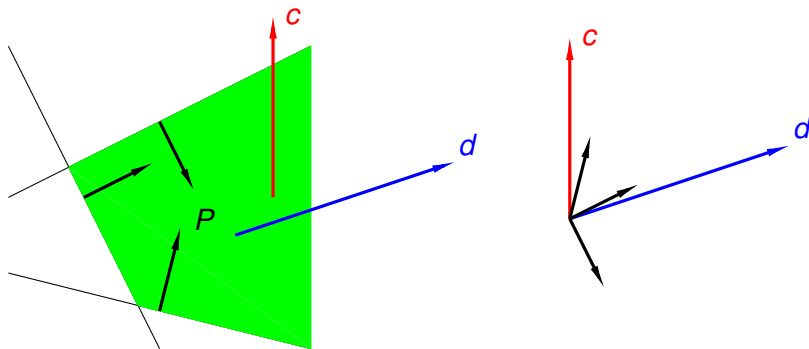
- 1 there is a ray d entirely inside the feasible region P , and
- 2 the objective c keeps increasing along the points on the ray.



An orthogonal wedge requires a value for M . How large should M be?

inward pointing normals

Consider the inward pointing normals to P .

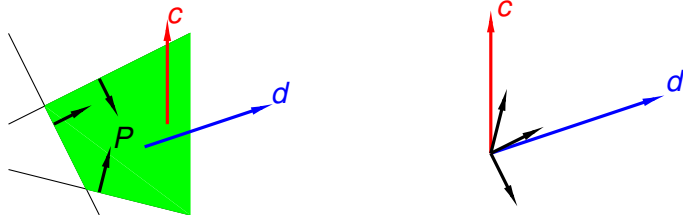


Compare the orientation of the inward pointing normals to the orientations of c and d .

inner normals and a criterion

Definition (inner normal)

The half plane $h: a_x x + a_y y \leq b$ has the *inner normal* $n_h = (-a_x, -a_y)$.



Lemma (criterion for unbounded LP)

The LP program defined by (H, c) is unbounded if and only if

- 1 $\bigcap_{h \in H} h \neq \emptyset$ and there exists a ray $d: d \cdot c > 0$, and
- 2 for all $h \in H: n_h \cdot d > 0$.

proof of the criterion

Lemma (criterion for unbounded LP)

The LP program defined by (H, c) is unbounded if and only if

- ① $\bigcap_{h \in H} h \neq \emptyset$ and there exists a ray $d: d \cdot c > 0$, and
- ② for all $h \in H: n_h \cdot d > 0$.

Proof. If and only if means $\Rightarrow$ and $\Leftarrow$.

$\Rightarrow$ If unbounded, then for the ray $d: d \cdot c > 0$ and $n_h \cdot d > 0$.

$\Leftarrow$ If feasible, take the point $p \in \bigcap_{h \in H}$, consider $p + \lambda d$.

$$d \cdot c > 0 \Rightarrow (p + \lambda d) \cdot c = p \cdot c + \lambda d \cdot c \text{ increases as } \lambda \rightarrow \infty.$$

$$n_h \cdot d > 0 \Rightarrow n_h(p + \lambda d) = \underbrace{n_h p}_{\geq b} + \lambda \underbrace{n_h \cdot d}_{> 0} > b, \text{ for } \lambda > 0.$$

Q.E.D.

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reduction to one dimension

The lemma gives an $O(n^2)$ algorithm.

We can do better if the objective $c = (0, 1)$, changing coordinates.

The ray d , such that $d \cdot c > 0$ can be normalized to $d = (d_x, 1)$.

For $n_h = (n_x, n_y)$, the condition $n_h \cdot d > 0$ then becomes

$$n_x d_x + n_y > 0 \quad \text{or equivalently} \quad n_x d_x > -n_y.$$

For all $h \in H$, we obtain linear inequalities in one variable d_x .

If this one dimensional linear program is infeasible,
then we have two half lines with an empty intersection.

certificates for unboundedness

For $n_h = (n_x, n_y)$, the condition $n_h \cdot d > 0$ then becomes

$$n_x d_x + n_y > 0 \quad \text{or equivalently} \quad n_x d_x > -n_y.$$

For all $h \in H$, we obtain linear inequalities in one variable d_x .

The corresponding half planes h_i and h_j are *certificates*, as h_i and h_j show that the linear programming problem is bounded. Take h_i and h_j as the first two half planes.

Exercise 5: Give a numerical example of an unbounded LP problem and compute a ray via reduction to one dimension.

a certificate for emptiness?

Exercise 6:

What is a certificate for a polygon defined by n half planes to be empty?

- 1 Consider the incremental linear programming problem and describe the relation with the certificate for emptiness problem.
- 2 Compare the cost of deciding whether a polygon is empty versus the cost of computing the intersection of the half planes.

Can the emptiness decision happen faster than the intersection computation?

outline of the algorithm 2DRANDOMIZEDLP

Algorithm 2DRANDOMIZEDLP(H, c)

Input: $H = (h_1, h_2, \dots, h_n)$, $c = (c_1, c_2)$.

Output: (x, y) optimal vertex if feasible, or
a ray if unbounded, or
two or more certificate half planes if infeasible.

- 1 Search for a ray d : $d \cdot c > 0$ and $d \cdot n_h > 0$ for all $h \in H$.
- 2 If the ray d exists then
- 3 if the half planes for which $d \cdot n_h = 0$ are feasible,
- 4 then return d as the ray showing unbounded LP,
- 5 else return with report that LP is infeasible.
- 6 Let h_1 and h_2 prove that (H, c) is bounded, $v_2 = \ell_1 \cap \ell_2$.
- 7 Run 2DRANDOMIZEDBOUNDEDLP on $(h_3, \dots, h_n)$.

cost of two dimensional linear programming

Theorem (cost of two dimensional linear programming)

A two dimensional linear programming problem with n constraints can be solved

- *in $O(n)$ randomized expected time*
- *using worst case $O(n)$ storage.*

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the LP solver of CGAL

A first example:

$$\begin{array}{llll} \text{minimize} & -32y + 64 & & \\ \text{subject to} & x + y & \leq & 7 \\ & -x + 2y & \leq & 4 \\ & x & \geq & 0 \\ & y & \geq & 0 \\ & y & \leq & 4 \end{array}$$

Copied from the CGAL documentation.

the LP solver of CGAL

The geometry of a first example:

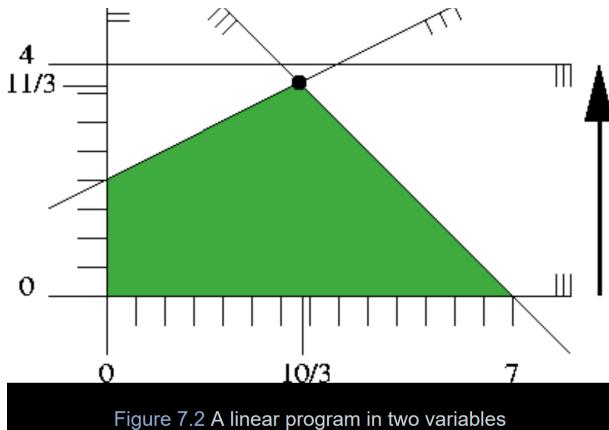


Figure 7.2 A linear program in two variables

Copied from the CGAL documentation.

defining a linear program

```
// by default, we have a nonnegative LP with  $Ax \leq b$ 
Program lp (CGAL::SMALLER, true, 0, false, 0);

// now set the non-default entries
const int X = 0;
const int Y = 1;
lp.set_a(X, 0, 1);
lp.set_a(Y, 0, 1);
lp.set_b(0, 7);           //  $x + y \leq 7$ 
lp.set_a(X, 1, -1);
lp.set_a(Y, 1, 2);
lp.set_b(1, 4);           //  $-x + 2y \leq 4$ 
lp.set_u(Y, true, 4);     //  $y \leq 4$ 
lp.set_c(Y, -32);         //  $-32y$ 
lp.set_c0(64);           //  $+64$ 

// solve the program, using ET as the exact type
Solution s = CGAL::solve_linear_program(lp, ET());
```

compiling and running

Compile and link with GMP:

```
$ g++ -o first_lp first_lp.cpp -lgmp
```

Running at the command prompt \$:

```
$ ./first_lp
status:          OPTIMAL
objective value: -160/3
variable values:
  0: 10/3
  1: 11/3
$
```

summary and exercises

The construction of the dual (or polar) of a polygon is taken from *Lectures on Polytopes* by G.M. Ziegler, Springer-Verlag, 1995.

We covered section 4.4 and 4.5 in the textbook.

Consider the following activities, listed below.

- 1 Write the solutions to the exercises.
- 2 Consider the exercises 13, 15, 16 in the textbook.